Introduction to Web Technologies (26CSE0106)

Topic : User Interface (UI) & User Experience (UX) Design

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Learning Outcomes

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After this lecture, students will be able to:

01
Foundations
Introduction to UI & UX, and why good design matters

02
The User centered deisgn
User engagement and user-centered design

03
Design Trade-offs
Intuitiveness versus efficiency

04
Design Principles
Core principles that guide UI design

05
Goals & Constraints
What design must achieve, and what limits it

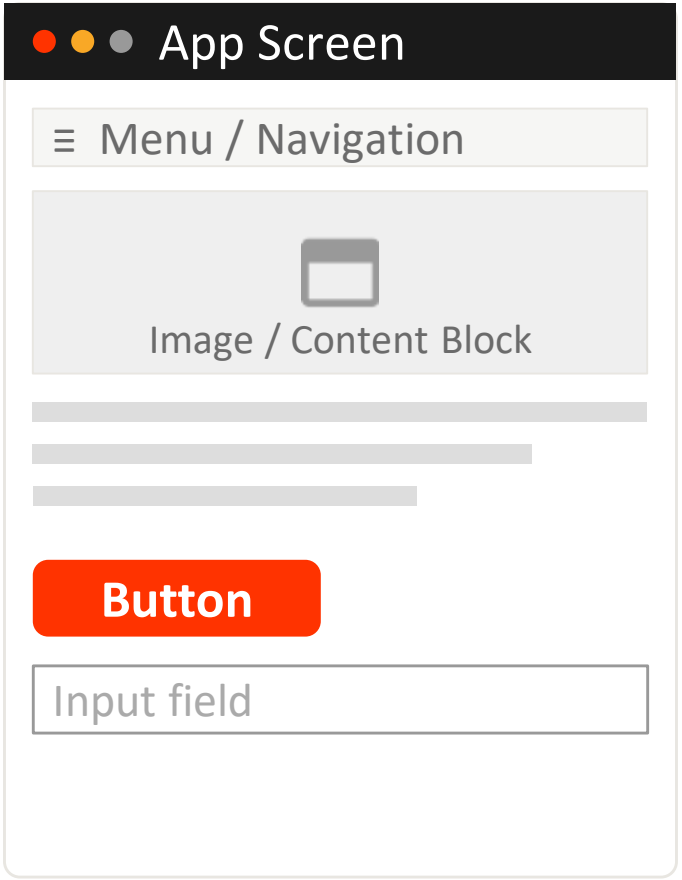
06
Modern Styles
Minimalist, flat, and navigation-based design

User Interface (UI)

The **User Interface (UI)** is the point of interaction between a human and a digital system — everything a user can see, touch, click, or type on screen.

- Buttons, icons & menus
- Colours, typography & imagery
- Layout & spacing
- Input controls & forms

UI is concerned with the LOOK and LAYOUT of a product.

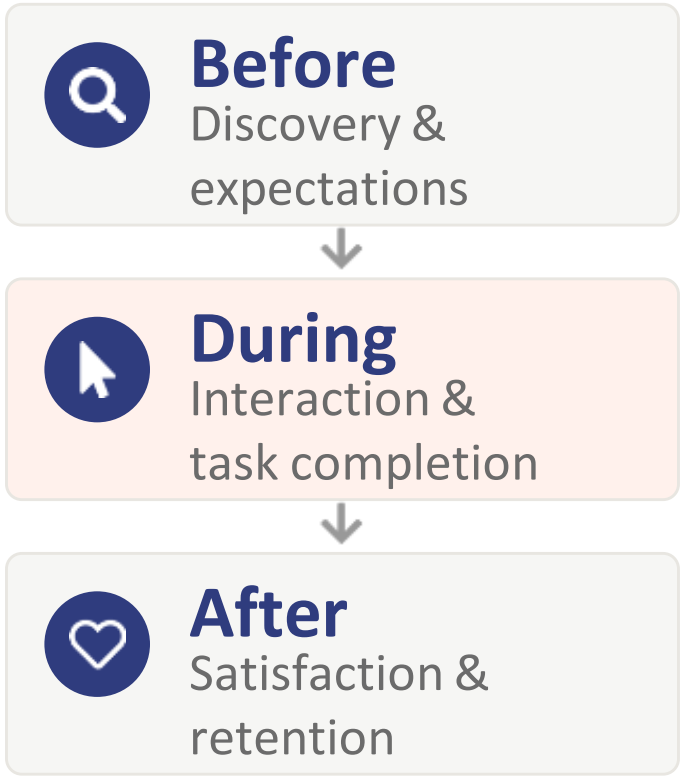


Every visible, clickable element = the UI

User Experience (UX)

User Experience (UX) is the overall feeling, satisfaction, and effectiveness a user experiences while interacting with a product — before, during, and after use.

- Ease of navigation & task flow
- Usability & cognitive load
- Satisfaction & trust
- Usefulness & accessibility



UX is concerned with the FEEL and FUNCTION of a product as a whole.

UI vs UX — Two Sides of One Product

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A useful analogy: if a product were a car, UI is the dashboard, seats, and steering wheel — UX is how it feels to drive.

UI — User Interface

- Visual design & aesthetics
- How the product looks
- Screens, buttons, icons, colour
- A single touchpoint
- Answers: “Is it attractive & clear?”

+

UX — User Experience

- Overall usability & feel
- How the product works
- Flows, structure, research, testing
- The entire journey
- Answers: “Is it useful & easy to use?”

Key idea: UI without UX is a beautiful screen that is hard to use. UX without UI is a logical flow that looks unpolished. Great products need both.

The Need for Good UI and UX

Design quality directly drives whether a product succeeds or is abandoned — it is not a cosmetic afterthought.

Builds Trust & Credibility

Reduces Friction

Improves Business Outcomes

Increases Accessibility

Drives Satisfaction & Loyalty

Prevents Costly Errors

Real-World Example: Poor UI/UX vs Good UI/UX

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Consider a course-registration portal used by thousands of students every semester:

⚠ Poor UI/UX

✗ Course search buried 4 menus deep
✗ No feedback after clicking "Submit"
✗ Errors shown only as red text, no explanation
✗ Inconsistent buttons across pages
✗ Result: students miss deadlines, flood the helpdesk

✔ Good UI/UX

✓ Search & filters visible on one screen
✓ Instant confirmation with clear next steps
✓ Errors explain exactly what to fix, inline
✓ Consistent, predictable button styles
✓ Result: faster registration, fewer support tickets

Takeaway: the same functionality, delivered with better design, changes real outcomes — time saved, errors avoided, and trust earned.

User Engagement

User engagement is the depth and quality of a user's interaction with a product — measured by attention, participation, and willingness to return.

What drives strong engagement:

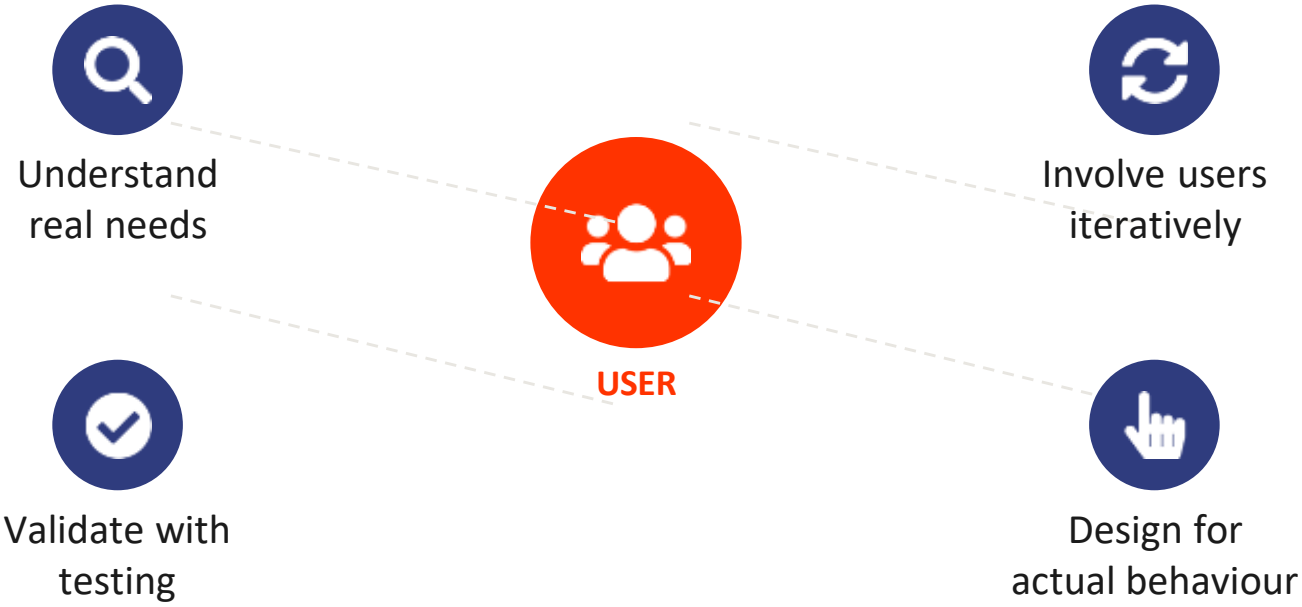
- **Relevance**
- **Responsiveness**
- **Feedback**
- **Simplicity**

How Engagement Is Measured

- **Session duration**
  Time actively spent per visit
- **Task completion rate**
  % of users who finish a flow
- **Return frequency**
  How often users come back
- **Feature adoption**
  Use of key product features
- **Drop-off points**
  Where users disengage or quit

User-Centered Design (UCD)

User-Centered Design is a design philosophy that places real user needs, behaviours, and feedback at the centre of every design decision — rather than assumptions or aesthetics alone.



Contrast: Designer-centered design relies on internal opinions or trends; user-centered design relies on evidence from real users.

Benefits of User-Centered Design

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- Higher Usability
- Greater Satisfaction
- Lower Development Cost
- Wider Accessibility
- Faster Adoption
- Reduced Risk

Intuitiveness versus Efficiency

Two desirable qualities that can pull interface design in different directions.



Intuitiveness

The quality of being usable without prior instruction — a new user can guess what to do correctly, on the first try.

Example: A trash-can icon for “delete” — understood instantly, with no manual.

Favours: familiarity, discoverability, minimal learning curve



Efficiency

The quality of enabling a task to be completed in the fewest possible steps, clicks, or amount of time — for an experienced user.

Example: Keyboard shortcuts (Ctrl+S) — fast for experts, invisible to beginners.

Favours: shortcuts, dense controls, expert-level speed

Intuitiveness versus Efficiency



Designing purely for beginners can slow down experts; designing purely for experts can confuse beginners. Good design finds the balance.

Real-World Example: Design Software

Microsoft Paint is highly intuitive — nearly anyone can open it and draw immediately. Adobe Photoshop is highly efficient for trained designers, with dense panels and shortcuts, but overwhelming for a first-time user. Neither is “wrong” — each is optimised for a different user and goal.

Principles of User Interface Design

These are widely accepted rules of thumb — grounded in cognitive psychology and decades of practice — that make interfaces easier to learn and use.



Clarity



Consistency



Visibility of Status



Feedback



Simplicity



Error Prevention



Flexibility & Efficiency



Accessibility

Core Principles I — Clarity, Consistency, Visibility

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Clarity

Every element's purpose should be immediately obvious — avoid ambiguity in labels, icons, and structure.

Example: A "Delete" button labelled clearly, not just a vague icon with no tooltip.



Consistency

Similar elements should look and behave the same way throughout the product, matching platform conventions.

Example: The same shade of blue always represents a clickable link across every screen.



Visibility of System Status

The system should always keep users informed about what is happening through timely, appropriate feedback.

Example: A progress bar during file upload shows percentage complete.

Core Principles II — Feedback, Simplicity, Error Prevention



Feedback

Every user action should trigger a clear, immediate response confirming what happened.

Example: A form field turns green with a checkmark after valid input.



Simplicity

Reduce unnecessary complexity; show only what the user needs at each step (minimise cognitive load).

Example: A checkout form asks only for essential fields, not every optional detail.



Error Prevention

Design should prevent mistakes before they happen, and make errors easy to recover from.

Example: A confirmation dialog before permanently deleting a file.

Core Principles III — Flexibility & Accessibility

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Flexibility & Efficiency of Use

Interfaces should serve both novice and expert users — e.g. through accelerators like shortcuts, without cluttering the experience for beginners.

Example: Gmail offers a full menu for new users, plus keyboard shortcuts ("c" to compose) for power users.

Accessibility

Design must be usable by people with diverse abilities — including visual, auditory, motor, and cognitive differences.

Example: Sufficient colour contrast, screen-reader labels (alt text), and full keyboard navigation.

Together, these eight principles form a practical checklist for evaluating any interface.

Design Goals — What Good Design Aims For

Usability goals define what a well-designed interface should achieve for its users:

Effectiveness
Users can accomplish their goals accurately & completely.

Efficiency
Tasks are completed with minimal time and effort.

Learnability
New users can quickly learn to use the system.

Memorability
Returning users easily recall how it works.

Error Tolerance
The system prevents, or gracefully recovers from, errors.

Satisfaction
The experience is pleasant and builds user confidence.

Design Constraints — What Limits the Design

Designers rarely have unlimited freedom. Real projects are shaped by practical constraints:

Technical
Device capability, screen size, bandwidth, and platform limitations.

Business
Budget, timelines, and return-on-investment expectations.

Time
Deadlines that limit how much research or iteration is possible.

User
Varying skill levels, languages, and contexts of use.

Accessibility & Legal
Compliance standards (e.g. WCAG) and regional regulations.

Platform
Conventions imposed by iOS, Android, web, or desktop environments.

Balancing Design Goals and Constraints

Good design is the art of achieving the best possible usability goals within real-world constraints — not ignoring either side.

Goals (what to achieve)

- Effectiveness
- Efficiency
- Satisfaction
- Learnability



Constraints (what limits it)

- Budget & timeline
- Device capability
- Legal / accessibility
- User diversity

Example: Accessibility guidelines (a constraint) often improve learnability and satisfaction (goals) — constraints and goals can reinforce each other, not just conflict.

Modern UI Design Styles — How We Got Here

UI visual style has evolved as screens, devices, and user expectations changed:



Skeuomorphism

UI mimicked real-world textures (leather, paper, buttons) to feel familiar.

Minimalism

Stripped away ornamentation; only essential elements remain.

Flat Design

Simple shapes, bold colour, and typography replace 3D effects entirely.

Navigation-Based

Structure and wayfinding become the primary design focus.

Minimalist Design

A style built on the principle “less is more” — removing every element that does not directly serve the user’s task.

Key characteristics:

- Generous white space between elements
- Limited colour palette (often monochrome + one accent)
- Only essential content and controls shown
- Strong reliance on typography hierarchy

Where you’ll see it: Google’s homepage, Apple product pages, Kindle reading apps.



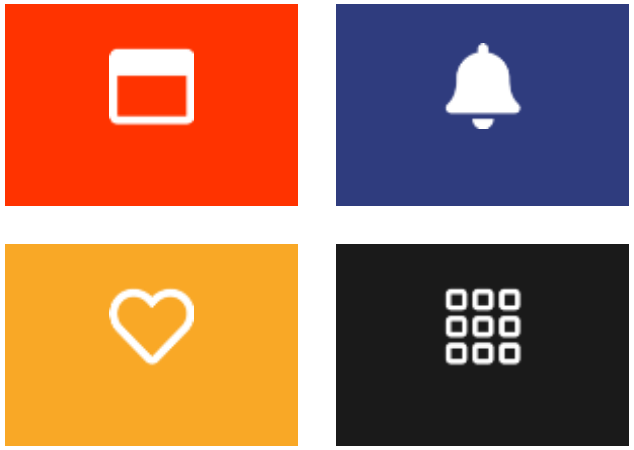
Flat Design

Interface elements are rendered as simple two-dimensional shapes, without gradients, shadows, or textures that mimic real-world depth.

Key characteristics:

- No skeuomorphic shadows, gradients, or bevels
- Bold, saturated blocks of solid colour
- Simple geometric icons and shapes
- Clean, sans-serif typography

Where you'll see it: Microsoft's "Metro"/Fluent design, Windows tiles, many mobile app icons.



Solid colour blocks, zero depth

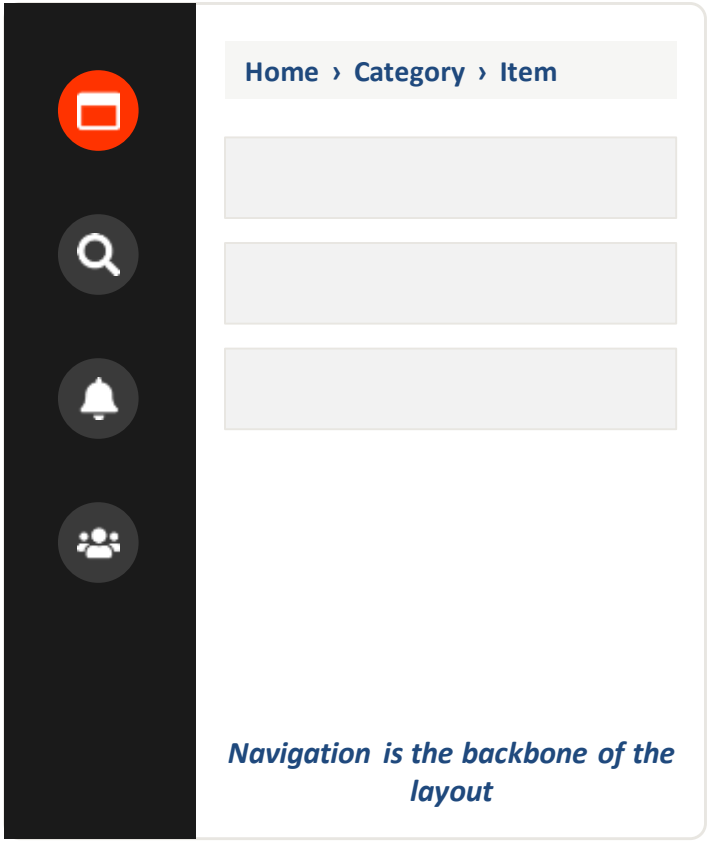
Navigation-Based Design

A style organised around how users move through the product — the navigation system itself becomes the primary structural and visual anchor.

Key characteristics:

- Persistent menus, tab bars, or side navigation
- Clear information architecture & wayfinding
- Breadcrumbs and progress indicators for orientation
- Prioritises findability over decorative styling

Where you'll see it: E-commerce category menus, mobile app bottom tab bars, dashboard sidebars.



Comparing the Three Styles

	Minimalist	Flat Design	Navigation-Based
Focus	Removing clutter	Flatness & colour	Structure & flow
Visual cues	White space, typography	Bold flat shapes, no depth	Menus, tabs, breadcrumbs
Best for	Content-first products	Brand-forward, modern apps	Complex, multi-section apps
Risk if overused	Can feel empty or unclear	Can reduce clickability cues	Can feel rigid or busy

These styles are not mutually exclusive — most production apps combine minimalist visuals with flat elements inside a strong navigation structure.

Choosing the Right Style for Your Product

Style is a means to an end — it must serve the design goals and users established earlier in this lecture, not personal preference or trends alone.

What is the product's complexity?

More screens & data → lean navigation-based.

Who are the users?

Novices favour intuitiveness & simplicity; experts tolerate density.

What is the core task?

Content consumption favours minimalism; multi-step tasks favour clear navigation.

What are the constraints?

Accessibility & platform guidelines may dictate colour contrast, spacing, and structure.

Discussion & In-Class Activity

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Activity: Heuristic Walkthrough

In pairs, open an app or website you use daily. Identify one example each of: clarity, consistency, and feedback — and one place where the design violates a principle from this lecture.

Discussion questions:

1. Can an interface be highly intuitive but poor UX overall? Give an example.
2. Which of the eight UI principles is hardest to apply consistently, and why?
3. Pick one app: is its style closer to minimalist, flat, or navigation-based? Justify your answer.
4. How might design constraints differ between a startup app and a government service?